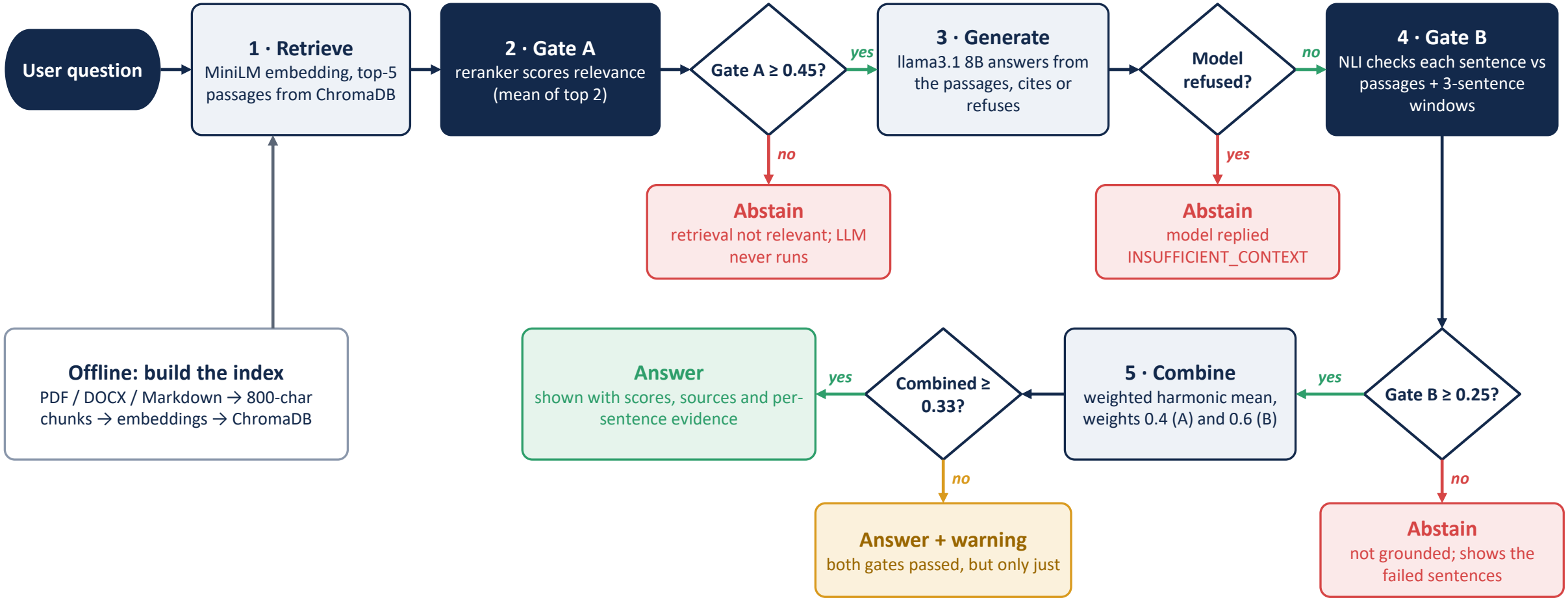
System workflow: what happens to one question

Two checks, four possible outcomes. Every outcome is logged.



Every outcome is logged: one SQLite row per question (scores, verdict, per-sentence labels, latency, thresholds) → Monitoring dashboard, and the benchmark and audit scripts.

Project workflow: five milestones

Build, measure, calibrate, stress-test, then audit and improve.

M1

Build

Architecture and pipeline: ingestion, retrieval, both gates, composite, telemetry, Streamlit app

31 unit tests; both gates add ~134 ms

→

M2

Benchmark

SQuAD 2.0 harness, 150 questions (half unanswerable), guarded vs plain RAG baseline

Hallucinations 75 → 3 of 75, but 84% over-abstention

→

M3

Calibrate

Data-fitted thresholds, risk–coverage analysis, NLI vs LLM-judge agreement study

Over-abstention 84% → 72%; judge agreement at chance

→

M4

Stress-test

HotpotQA multi-hop questions; sentence vs 800-character chunk indexing

Full recall 31% → 66%; Gate B weak on multi-hop

→

M5

Audit & improve

Re-scored every result, fixed the scoring errors, built and tested Gate B v2, final deliverables

Over-abstention 71% → 48% ($p < 0.001$)

M5 went back over every saved result from M2–M4: corrected the scoring, counted how many refusals came from the prompt versus the gates, and reduced Gate B's false rejections.

Deliverables: working app with guide and dashboard · 92 tests · 5 milestone reports · A3 poster · project report · presentation